

LIMITATIONS OF THE CAPM

1. The model **makes unrealistic assumptions**
2. The **parameters of the model cannot be estimated precisely**

The market index used can be wrong.

The firm may have changed during the 'estimation' period'

3. The **model does not work well**

- If the model is right, there should be:

 - A linear relationship between returns and betas

 - The only variable that should explain returns is betas

- The reality is that

 - The relationship between betas and returns is weak

 - Other variables (size, price/book value) seem to explain differences in returns better.

ALTERNATIVES TO THE CAPM

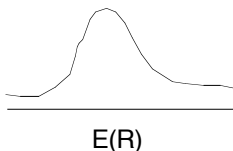
Step 1: Defining Risk

The risk in an investment can be measured by the variance in actual returns around an expected return

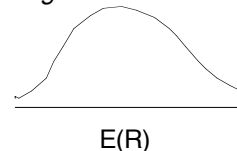
Riskless Investment



Low Risk Investment



High Risk Investment



Step 2: Differentiating between Rewarded and Unrewarded Risk

Risk that is specific to investment (Firm Specific)

Can be diversified away in a diversified portfolio
 1. each investment is a small proportion of portfolio
 2. risk averages out across investments in portfolio

The marginal investor is assumed to hold a “diversified” portfolio. Thus, only market risk will be rewarded and priced.

Risk that affects all investments (Market Risk)

Cannot be diversified away since most assets are affected by it.

Step 3: Measuring Market Risk

The CAPM	The APM	Multi-Factor Models	Proxy Models
If there is 1. no private information 2. no transactions cost the optimal diversified portfolio includes every traded asset. Everyone will hold this market portfolio Market Risk = Risk added by any investment to the market portfolio:	If there are no arbitrage opportunities then the market risk of any asset must be captured by betas relative to factors that affect all investments. Market Risk = Risk exposures of any asset to market factors	Since market risk affects most or all investments, it must come from macro economic factors. Market Risk = Risk exposures of any asset to macro economic factors.	In an efficient market, differences in returns across long periods must be due to market risk differences. Looking for variables correlated with returns should then give us proxies for this risk. Market Risk = Captured by the Proxy Variable(s)
Beta of asset relative to Market portfolio (from a regression)	Betas of asset relative to unspecified market factors (from a factor analysis)	Betas of assets relative to specified macro economic factors (from a regression)	Equation relating returns to proxy variables (from a regression)

WHY THE CAPM PERSISTS...

- The CAPM, notwithstanding its many critics and limitations, has **survived as the default model** for risk in equity valuation and corporate finance.
- The alternative models that have been presented as better models (APM, Multifactor model..) have made inroads in performance evaluation but not in prospective analysis because:
 - The alternative models (which are richer) **do a much better job than the CAPM in explaining past returns**, but their effectiveness drops off when it comes to estimating expected future returns (because the models tend to shift and change).
 - The alternative models are **more complicated and require more information** than the CAPM.
 - For most companies, the **expected returns you get with the the alternative models is not different enough** to be worth the extra trouble of estimating four additional betas.

APPLICATION TEST: WHO IS THE MARGINAL INVESTOR IN YOUR FIRM?

- You can get information on insider and institutional holdings in your firm from:
 - <http://finance.yahoo.com/>
 - Enter your company's symbol and choose profile.
- Looking at the breakdown of stockholders in your firm, consider whether the marginal investor is
 - An institutional investor
 - An individual investor
 - An insider
- Follow up by evaluating whether the marginal investor is likely to be diversified.
 - If yes, you are on safer ground using the risk and return models that assume that only non-diversifiable risk is rewarded.
 - If no, you will have to adapt your risk measure to bring in some or all of the company-specific risk that you were ignoring.

FROM RISK MODELS TO HURDLE RATES: ESTIMATION CHALLENGES

“The price of purity is
purists...”

INPUTS REQUIRED TO USE THE CAPM -

- The capital asset pricing model yields the following expected return:
 - $\text{Expected Return} = \text{Riskfree Rate} + \text{Beta} * (\text{Expected Return on the Market Portfolio} - \text{Riskfree Rate})$
- To use the model, we need three inputs:
 - The **current risk-free rate**
 - The **expected equity risk premium**, the premium expected for investing in risky assets, i.e. the market portfolio, over the riskless asset.
 - The **beta of the asset** being analyzed.

THE RISKFREE RATE AND TIME HORIZON

- On a riskfree asset, **the actual return is always equal to the expected return**. Therefore, there is no variance around the expected return.
- For an investment to be riskfree, i.e., to have an actual return be equal to the expected return, two conditions have to be met –
 - There **can be no default risk**, which generally implies that the security has to be issued by the government. Note, however, that not all governments can be viewed as default free.
 - There **can be no uncertainty about reinvestment rates**, which implies that it is a zero-coupon security with the same maturity as the cash flow being analyzed.

RISKFREE RATE IN PRACTICE

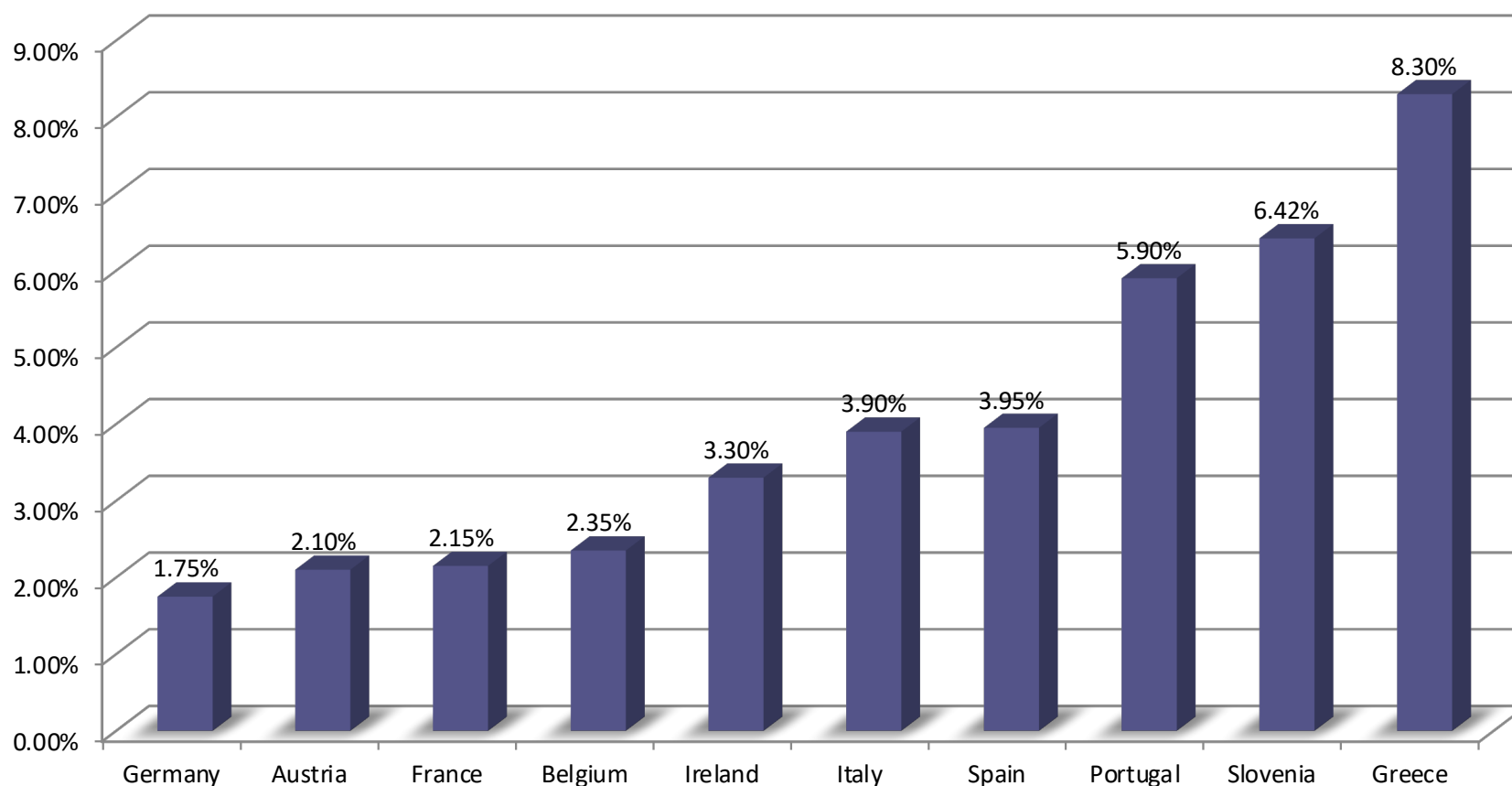
- **Definition:** The riskfree rate is the **rate on a zero coupon default-free bond** matching the time horizon of the cash flow being analyzed.
- **Implication:** Theoretically, this translates into **using different riskfree rates for each cash flow** - the 1 year zero coupon rate for the cash flow in year 1, the 2-year zero coupon rate for the cash flow in year 2 ...
- **A Practical Solution:** Practically speaking, if there is substantial uncertainty about expected cash flows, the present value effect of using time varying riskfree rates is small enough that it may not be worth it.
- **In corporate finance, almost everything we do is long term. So, using a long-term default free rate as the riskfree rate makes sense.**

THE BOTTOM LINE ON RISKFREE RATES

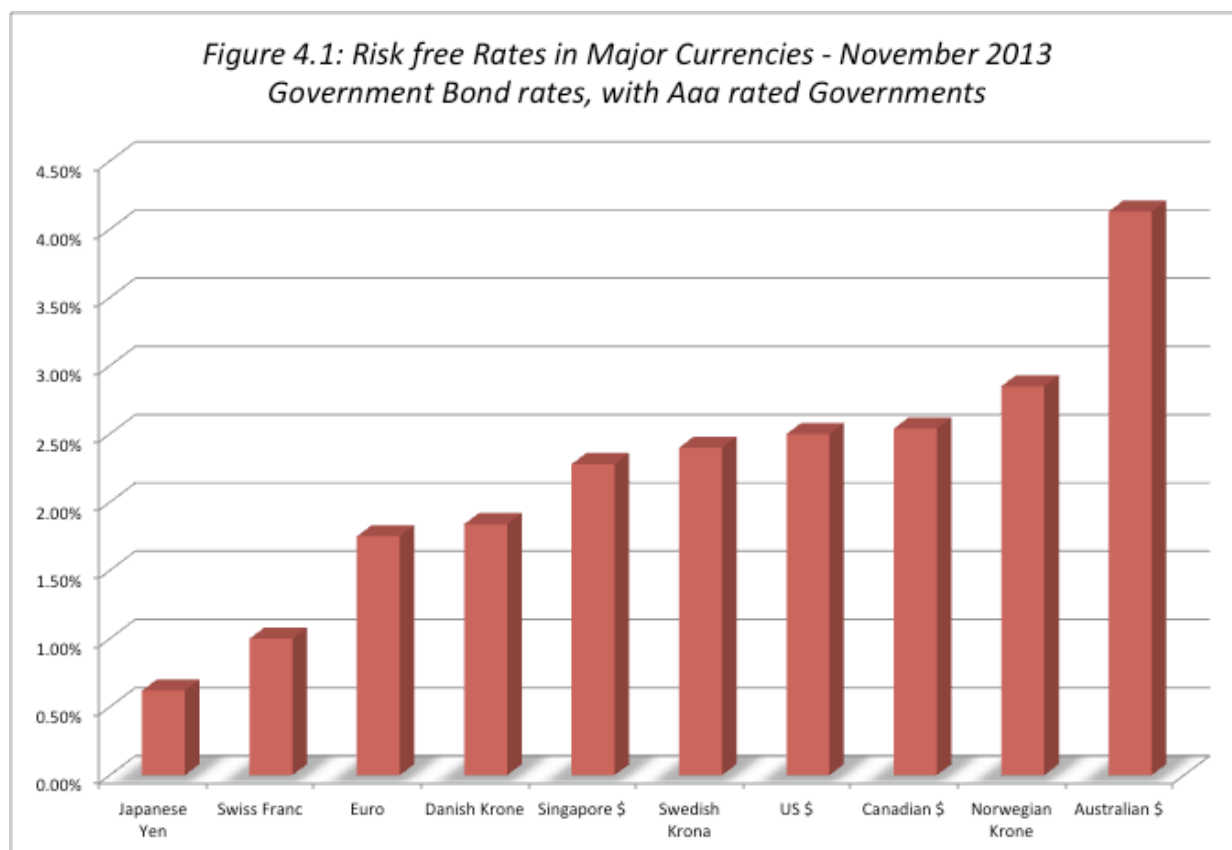
- **Currency Matching:** The riskfree rate that you use in an analysis should be in the same currency that your cashflows are estimated in.
 - In other words, if your cashflows are in U.S. dollars, your riskfree rate has to be in U.S. dollars as well.
 - If your cash flows are in Euros, your riskfree rate should be a Euro riskfree rate.
- **Just use the government bond rate?** The conventional practice of estimating riskfree rates is to use the government bond rate, with the government being the one that is in control of issuing that currency.
- **If the government is default-free,** using a long term government rate (even on a coupon bond) as the risk free rate on all of the cash flows in a long term analysis will yield a close approximation of the true value. With US dollars in November 2013, for instance, the ten-year treasury bond rate of 2.75% was used as the riskfree rate.

WHAT IS THE EURO RISKFREE RATE? AN EXERCISE IN NOVEMBER 2013

Rate on 10-year Euro Government Bonds: November 2013



WHEN THE GOVERNMENT IS DEFAULT FREE: RISK FREE RATES — IN NOVEMBER 2013



WHAT IF THERE IS NO DEFAULT-FREE ENTITY?

RISK FREE RATES IN NOVEMBER 2013

- **Adjust the local currency government borrowing rate for default risk** to get a riskless local currency rate.
 - In November 2013, the Indian government rupee bond rate was 8.82%. the local currency rating from Moody's was Baa3 and the default spread for a Baa3 rated country bond was 2.25%.

$$\text{Riskfree rate in Rupees} = 8.82\% - 2.25\% = 6.57\%$$

- In November 2013, the Chinese Renmimbi government bond rate was 4.30% and the local currency rating was Aa3, with a default spread of 0.8%.

$$\text{Riskfree rate in Chinese Renmimbi} = 4.30\% - 0.80\% = 3.50\%$$

- **Do the analysis in an alternate currency**, where getting the riskfree rate is easier. With Vale in 2013, we could choose to do the analysis in US dollars (rather than estimate a riskfree rate in R\$). The riskfree rate is then the US treasury bond rate.
- **Do your analysis in real terms**, in which case the riskfree rate has to be a real riskfree rate. The inflation-indexed treasury rate is a measure of a real riskfree rate.

THREE PATHS TO ESTIMATING SOVEREIGN DEFAULT SPREADS

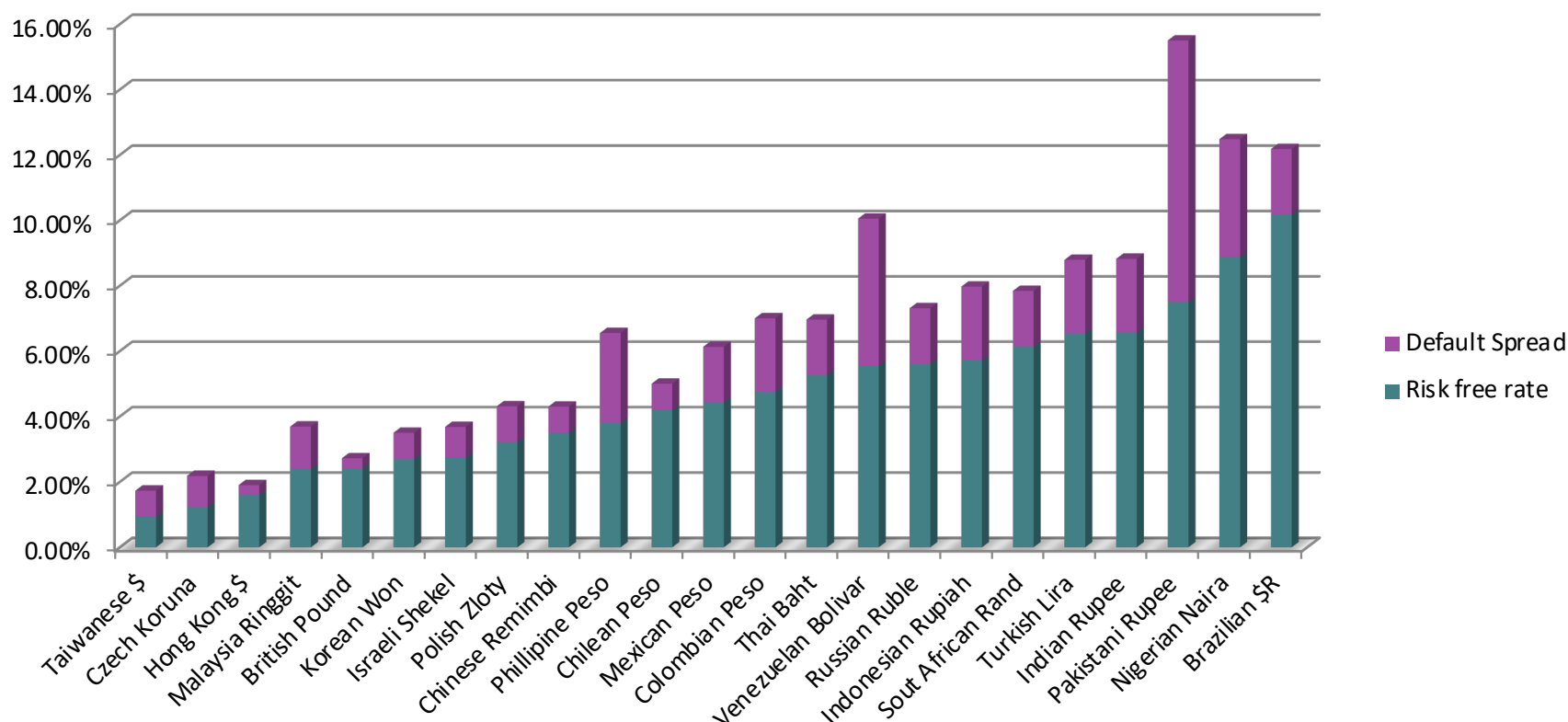
- **Sovereign dollar or euro denominated bonds:** The difference between the interest rate on a sovereign US \$ bond, issued by the country, and the US treasury bond rate can be used as the default spread. For example, in November 2013, the 10-year Brazil US \$ bond, denominated in US dollars had a yield of 4.25% and the US 10-year T.Bond rate traded at 2.75%.

$$\text{Default spread} = 4.25\% - 2.75\% = 1.50\%$$

- **Sovereign CDS spreads:** Obtain the default spreads for sovereigns in the CDS market. The CDS spread for Brazil in November 2013 was 2.50%.
- **Ratings-based spread:** If you know the sovereign rating for a country, you can estimate the default spread based on the rating. In November 2013, Brazil's rating was Baa2, yielding a default spread of 2%.

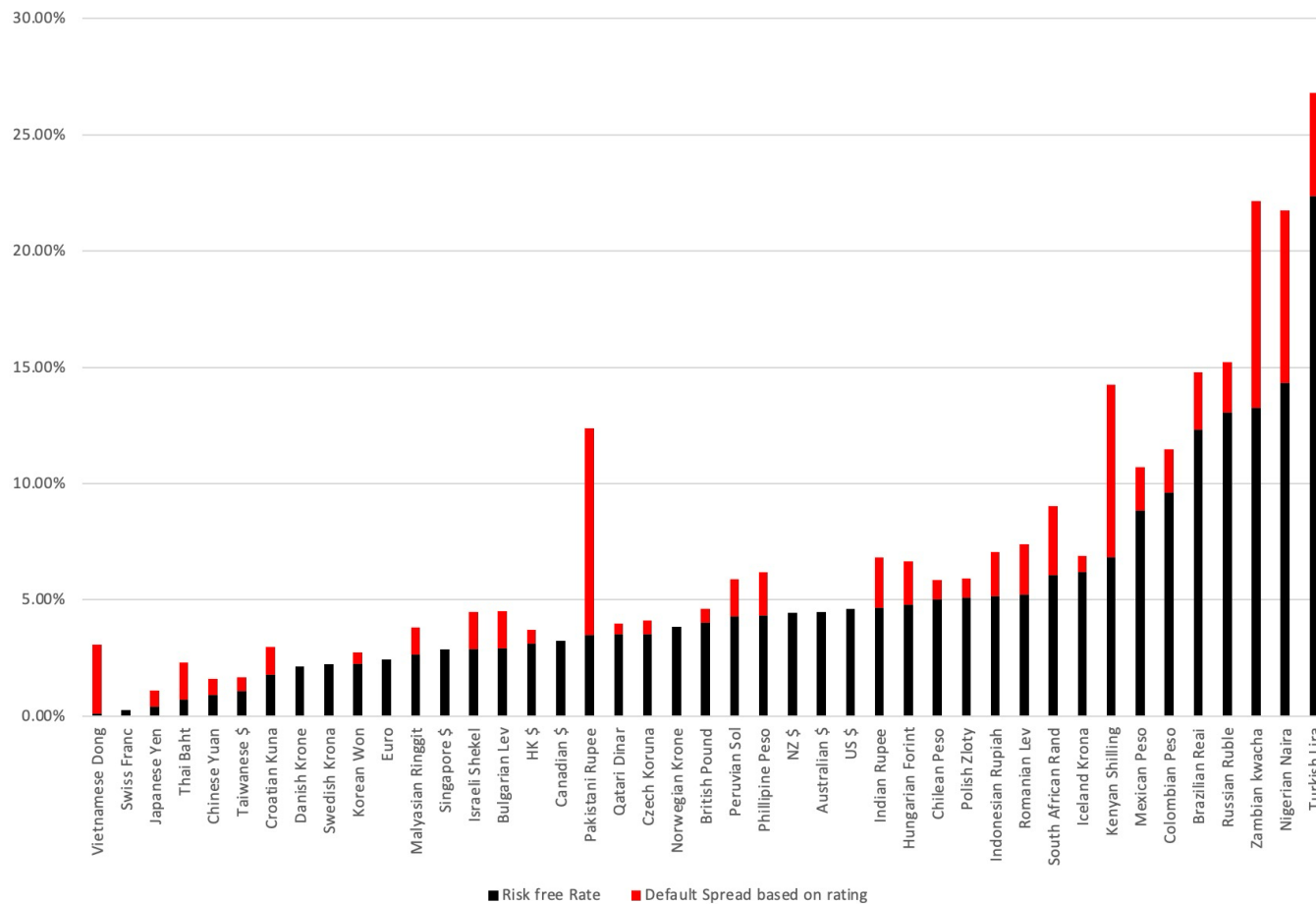
RISK FREE RATES IN CURRENCIES: SOVEREIGNS WITH DEFAULT RISK IN NOVEMBER 2013

Figure 4.2: Risk free rates in Currencies where Governments not Aaa rated



RISK FREE RATES IN JANUARY 2025

Government-bond Based Riskfree Rates in January 2025



MEASUREMENT OF THE EQUITY RISK PREMIUM

- The equity risk premium is **the premium that investors demand for investing in an average risk equity**, relative to the riskfree rate. In short, it is the price of risk in equity markets, rising with fear.
- As a general proposition, this premium should be
 - Greater than zero
 - Increase with the **risk aversion** of the investors in that market
 - Increase with the **riskiness of the “average” risk investment**
- If so, it also follows that equity risk **premiums should change over time**, as economic circumstances change and investor composition also changes.

WHAT IS YOUR RISK PREMIUM?

- Assume that stocks are the only risky assets and that you are offered two investment options:
 - a **riskless investment** (say a Government Security), on which you can make 3%
 - **An index fund of all stocks**, on which the returns are uncertain
- How much of an expected return would you demand to shift your money from the riskless asset to the mutual fund?
 - a. Less than 3%
 - b. Between 3% - 5%
 - c. Between 5% - 7%
 - d. Between 7% -9%
 - e. Between 9%- 11%
 - f. More than 11%

RISK AVERSION AND RISK PREMIUMS

- If this were the entire market, the risk premium would be a **weighted average of the risk premiums demanded** by each and every investor.
- The weights will be determined by the **wealth that each investor brings to the market**. Thus, Warren Buffett's risk aversion counts more towards determining the "equilibrium" premium than yours' and mine.
- As investors become more risk averse, or the market becomes "more risky", you would expect the "equilibrium" premium to increase.

RISK PREMIUMS DO CHANGE..

- Go back to the previous question. Assume now that you are making the same choice but that you are making it in the aftermath of a stock market crash (it has dropped 25% in the last month). Would you change your answer?
 - a. I would demand a larger premium
 - b. I would demand a smaller premium
 - c. I would demand the same premium
- If your equity risk premium rises, what should happen to stock prices, all else held constant?

ESTIMATING RISK PREMIUMS IN PRACTICE

- **Survey Premiums:** Survey investors on their desired risk premiums and use the average premium from these surveys.
- **Historical Premiums:** Assume that the actual premium delivered over long time periods is equal to the expected premium - i.e., use historical data.
- **Implied Premiums:** Estimate a forward-looking premium, based upon today's asset prices.

1. THE SURVEY APPROACH

- Surveying all investors in a marketplace is impractical.
- However, you can survey a few individuals and use these results. In practice, this translates into surveys of the following:

<i>Group Surveyed</i>	<i>Survey done by</i>	<i>Estimated ERP</i>	<i>Notes</i>
Individual Investors	Securities Industries Association	8.3% (2004)	One year premium
Institutional Investors	Merrill Lynch	4.8% (2013)	Monrthly updates
CFOs	Campbell Harvey & Graham	4.48% (2012)	5-8% response rate
Analysts	Pablo Fernandez	5.0% (2011)	Lowest standard deviation
Academics	Pablo Fernandez	5.7% (2011)	Higher for emerging markets

- The limitations of this approach are:
 - There are **no constraints on reasonability** (the survey could produce negative risk premiums or risk premiums of 50%)
 - The survey results are **more reflective of the past than the future**.
 - They **tend to be short term**; even the longest surveys do not go beyond one year.

2. THE HISTORICAL PREMIUM APPROACH

- This is the default approach used by most to arrive at the premium to use in the model
- In most cases, this approach does the following
 - Defines a **time period** for the estimation (1928-Present, last 50 years...)
 - Calculates **average returns on a stock index** during the period
 - Calculates **average returns on a riskless security** over the period
 - Calculates the difference between the two averages and uses it as a premium looking forward.
- The limitations of this approach are:
 - it assumes that the risk aversion of investors **has not changed in a systematic way across time**. (The risk aversion may change from year to year, but it reverts back to historical averages)
 - it assumes that **the riskiness of the “risky” portfolio (stock index) has not changed** in a systematic way across time.

HISTORICAL ERP: A HISTORICAL SNAPSHOT

	<i>Arithmetic Average</i>		<i>Geometric Average</i>	
	Stocks - T. Bills	Stocks - T. Bonds	Stocks - T. Bills	Stocks - T. Bonds
1928-2024	8.44%	7.00%	6.63%	5.44%
Std Error	<i>2.01%</i>	<i>2.12%</i>		
1975-2024	9.25%	7.03%	8.02%	6.22%
Std Error	<i>2.30%</i>	<i>2.67%</i>		
2015-2024	12.34%	13.54%	11.22%	12.71%
Std Error	<i>5.04%</i>	<i>3.84%</i>		

Historical
premium for
the US

- If you are going to use a historical risk premium, make it
 - Long term (because of the standard error)
 - Consistent with your choice of risk free rate
 - A “compounded” average
- No matter which estimate you use, recognize that it is backward looking, is noisy and may reflect selection bias.

3. A FORWARD-LOOKING ERP

- If you know **the price paid for an asset and have estimates of the expected cash flows on the asset**, you can estimate the **IRR** of these cash flows. If you paid the price, this is your expected return.
 - In the **bond market**, that is exactly what we do when we compute the yield to maturity on a bond.
 - If you assume that **stocks are correctly priced in the aggregate and you can estimate the expected cashflows from buying stocks, you can estimate the expected rate of return on stocks** by finding that discount rate that makes the present value equal to the price paid.
- Subtracting out the riskfree rate should yield **an implied equity risk premium**. This implied equity premium is a forward-looking number and can be updated as often as you want.

IMPLIED ERP IN NOVEMBER 2013: WATCH WHAT I PAY, NOT WHAT I SAY..

- If you can observe what investors are willing to pay for stocks, you can back out an expected return from that price and an implied equity risk premium.

Base year cash flow (last 12 mths)

Dividends (TTM): 33.22
 + Buybacks (TTM): 49.02
 = Cash to investors (TTM): 82.35
 Earnings in TTM:

Expected growth in next 5 years

Top down analyst estimate of earnings growth for S&P 500 with stable payout: 5.59%

Beyond year 5

Expected growth rate =
 Riskfree rate = 2.55%
 Expected CF in year 6 =
 108.1(1.0255)

E(Cash to investors) 86.96 91.82 96.95 102.38 108.10

S&P 500 on 11/1/13= 1756.54

$$1756.54 = \frac{86.96}{(1+r)} + \frac{91.82}{(1+r)^2} + \frac{96.95}{(1+r)^3} + \frac{102.38}{(1+r)^4} + \frac{108.10}{(1+r)^5} + \frac{110.86}{(r - .0255)(1+r)^5}$$

r = Implied Expected Return on Stocks = 8.04%

Minus

Risk free rate = T.Bond rate on 1/1/14=2.55%

Equals

Implied Equity Risk Premium (1/1/14) = 8.04% - 2.55% = 5.49%

THE BOTTOM LINE ON EQUITY RISK PREMIUMS IN NOVEMBER 2013

- **Mature Markets:** In November 2013, the number that we chose to use as the equity risk premium for all mature markets was 5.5%. This was set equal to the implied premium at that point in time and it was much higher than the historical risk premium of 4.20% prevailing then.

	Arithmetic Average		Geometric Average	
	Stocks - T. Bills	Stocks - T. Bonds	Stocks - T. Bills	Stocks - T. Bonds
1928-2012	7.65%	5.88%	5.74%	4.20%
	2.20%	2.33%		
1962-2012	5.93%	3.91%	4.60%	2.93%
	2.38%	2.66%		
2002-2012	7.06%	3.08%	5.38%	1.71%
	5.82%	8.11%		

- **For emerging markets,** we will use the melded default spread approach (where default spreads are scaled up to reflect additional equity risk) to come up with the additional risk premium that we will add to the mature market premium. Thus, markets in countries with lower sovereign ratings will have higher risk premiums than 5.5%.

$$\text{Emerging Market ERP} = 5.5\% + \text{Country Default Spread} * \left(\frac{\sigma_{\text{Equity}}}{\sigma_{\text{Country Bond}}} \right)$$

WHAT ABOUT EQUITY RISK PREMIUMS FOR OTHER MARKETS?

- Historical data for markets outside the United States is available **for much shorter time periods**. The problem is even greater in emerging markets.
- The historical premiums that emerge from this data reflects this data problem and there is **much greater error associated with the estimates of the premiums**.
- You could **try to compute implied equity risk premiums** but getting the inputs, especially for long term growth are difficult to do.

ONE SOLUTION: BOND DEFAULT SPREADS AS CRP — NOVEMBER 2013

- In November 2013, the equity risk premium for the US was 5.50% Using the default spread on the sovereign bond or based upon the sovereign rating and adding that spread to the mature market premium (4.20% for the US) gives you a total ERP for a country.

Country	Rating	Default Spread (Country Risk Premium)	US ERP	Total ERP for country
India	Baa3	2.25%	5.50%	7.75%
China	Aa3	0.80%	5.50%	6.30%
Brazil	Baa2	2.00%	5.50%	7.50%

- If you prefer CDS spreads:

<i>Country</i>	<i>Sovereign CDS Spread</i>	<i>US ERP</i>	<i>Total ERP for country</i>
India	4.20%	5.50%	9.70%
China	1.20%	5.50%	6.70%
Brazil	2.59%	5.50%	8.09%

BEYOND THE DEFAULT SPREAD? EQUITIES ARE RISKIER THAN BONDS

- While default risk spreads and equity risk premiums are highly correlated, one would expect equity spreads to be higher than debt spreads. One approach to scaling up the premium is to look at the relative volatility of equities to bonds and to scale up the default spread to reflect this:

$$\text{Country Risk Premium} = \text{Country Default Spread} * \left(\frac{\sigma_{\text{Equity}}}{\sigma_{\text{Country Bond}}} \right)$$

- Brazil: The annualized standard deviation in the Brazilian equity index over the previous year is 21 percent, whereas the annualized standard deviation in the Brazilian C-bond is 14 percent.
 - Brazil's Equity Risk Premium = 5.50% + 2.00% (21%/14%) = 8.50%
- Using the same approach for China and India:
 - China's Equity Risk Premium = 5.50% + 0.80% (18%/10%) = 6.94%
 - India's Equity Risk Premium = 5.50% + 2.25% (24%/17%) = 9.10%